CHENG XIANYUN

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EDUCATION

National University of Singapore (NUS) **Sep 2026 - Present**
Programme: Master of Science (Mechanical Engineering).

Central South University (CSU) & University of Dundee (UoD) **Sep 2022 - Jun 2026**
Dundee International Institute

- Degrees: BEng Mechanical Design, Manufacturing and Automation (CSU); BEng (Hons) Mechanical Engineering, First Class Honours (UoD).
- GPA: 86.33/100.
- Awards: Third-Class Scholarship (2022–2023) and Special Scholarship (2025), Central South University.

RESEARCH & PROJECT EXPERIENCE

Member **Oct 2025 - Present**
Research Team of Tianlin Li (Beihang University)

- Built a lightweight test wrapper and automated pipeline for LLM document workflows in high-stakes decision tasks, unifying input routing, format normalisation and model execution.
- Defined 3 metamorphic relations and ran 48,000 executions (4 LLM workflows × 4 tasks × 4 formats), using Jensen–Shannon divergence to quantify format-induced output and evidence shifts.
- Developed Vote Aggregation and Format Routing mitigations, reducing format-induced violations by up to 44.21%; published as second author at ASE 2026 (CCF-A), with a related patent application in progress.

Member **Apr 2026 - Present**
Research Team of Yang Liu (NTU Singapore)

- Integrated open-source AI agents into a modular evaluation framework, collecting multimodal interaction data including screenshots, actions, and execution traces while separating execution and assessment modules.
- Curated reviews spanning 9 game genres and 2 reviewer types, extracting review aspects, attitudes and evidence against shared and genre-specific evaluation criteria.
- Configured persona-conditioned reviewer agents with a consistency protocol comparing 3 dimensions, evidence grounding, review focus and ratings, against experienced human players.
- Related research papers have been submitted and are under double-blind review.

Team Member — TikTok TechJam 2026 **Aug 2026 – Sep 2026**
IntentCompass: Multi-turn Conversational Product Search Agent

- Analysed intent updates, retrieval behaviour and edge-case failures of a multi-turn product search agent over a 50,000-item catalogue, covering routing, constraints and clarification.
- Independently validated system reproducibility on Windows, conducted failure analysis and evaluation checks, and organised evaluation evidence and the final demo workflow.

Member **Feb 2025 - Apr 2025**
UV Photodetector Design and Process Optimisation

- Specified a 3-layer Ni/n-ZnO/Ni ultraviolet photodetector with a 3-stage workflow spanning fabrication, electrical characterisation and optical validation, linking process variables to device performance.
- Fabricated thin-film devices through RF magnetron sputtering, UV lithography and ultrasonic lift-off, gaining hands-on experience in deposition, patterning and device formation.
- Analysed defects associated with a 440 nm cutoff-wavelength shift and proposed sputtering and annealing adjustments to reduce defect-related spectral drift.

Member **Sep 2024 - Jan 2025**

Intelligent Firefighting Vehicle Design

- Modelled 3 mechanical subsystems, chassis, tracked drive and fire-suppression module, in SolidWorks and verified stress and deformation in ANSYS under representative loads.
- Engineered a PCB integrating 6 functional modules for motor control, power conversion, flame sensing, Bluetooth communication, relays and actuator interfaces across manual and automatic modes.
- Implemented Mbed PWM speed control, steering, sensor interrupts and fan/pump drive, then validated Bluetooth remote driving, flame detection and suppression on the prototype.

Member **Feb 2024 - Jun 2024**

Undergraduate Innovation and Entrepreneurship Training Programme - Lightweight Virtual Try-On System

- Combined 3 vision components, MoveNet, MobileNetV2 and FPN, for real-time human keypoint extraction and pose-aware garment alignment without dedicated GPU hardware.
- Streamlined CP-VTON+ through 2 architectural modifications, depth-wise separable convolution and garment-mask input, reducing computational complexity while preserving core virtual try-on functionality.
- Applied 2 deployment optimisations, white-garment segmentation refinement and OpenVINO CPU acceleration, enabling GPU-free inference across 2 platforms, Web and PC.

INTERNSHIP EXPERIENCE

Intern **Jan 2026 - Mar 2026**

Artificial Intelligence Institute, Platform and Engineering Department, CAICT

- Researched 2 regional data ecosystems, exploring high-quality dataset development in Ningbo and data annotation practices in Shenyang, and synthesised findings for internal reporting.
- Prepared agendas and reviewed materials for 2 technical events, including WG2/WG5 joint meetings and the Fangsheng Intelligent Testing Seminar, facilitating meeting preparation and technical communication.

Intern, Common Technology Research Institute **Oct 2025 - Dec 2025**

Changsha Jingjia Microelectronics Co., Ltd.

- Supported semiconductor process analysis, product testing and engineering data processing, analysing production and test data to flag potential process abnormalities and improvements.
- Trained in the packaging laboratory and production-line workflows, learning how backend manufacturing, integrated testing, quality control and process flow connect.

STUDENT ACTIVITIES

Volunteer, 11th ICAMMT **Apr 2025**

- Managed presenter logistics and on-site coordination for the 11th ICAMMT, assisting speakers throughout conference activities.
- Served as an interpreter and opening-ceremony emcee, facilitating communication between organisers, presenters and attendees.

Publicity Officer, Youth League Committee, Dundee International Institute **Sep 2022 - Sep 2023**

- Wrote and edited 20+ WeChat articles, producing publicity content for student activities.
- Organised practical-skills activities, including a Photoshop workshop, covering event preparation and participant coordination.

SKILLS & CERTIFICATIONS

- English: IELTS 6.5.
- Technical: Python, C++, MATLAB, SolidWorks, AutoCAD, ANSYS, Arduino, and MS Office.